King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Final Exam Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Evaluate

$$\int_1^2 \frac{1+x-x^2}{x^2+x^3}\,dx.$$

(a) $\frac{3}{2} - \ln 2$

(b) $\frac{1}{2} + \ln \frac{3}{2}$

(c) $\frac{1}{2} - \ln \frac{3}{2}$

(d) $\frac{3}{2} + \ln 2$

(e) $\frac{2}{3} - \ln 2$

2. Consider the convergent sequence $\{a_n\}$, where $a_1 = \sqrt{6}$, $a_{n+1} = \sqrt{6+a_n}$ for $n \geq 1$. The sequence converges to

(a) 6

(b) 3

(c) 1

(d) -1

(e) -2

3. The least number of terms required to approximate the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n!}$ with an error of less than 0.001 is

(a) 5

(b) 7

(c) 6

(d) 4

(e) 3

4. Evaluate

$$\int_0^2 \frac{dx}{(x-1)^2}.$$

(a) converges to 2

(b) converges to 1

(c) converges to -1

(d) converges to -2

(e) diverges

5. Evaluate

$$\int_0^{\pi^2} \sin\sqrt{x}\, dx.$$

(a) 6π

(b) 4π

(c) 2π

(d) 8π

(e) π

6.

$$\sum_{n=1}^{\infty} \left(e^{-n} + e^{-2n}\right) =$$

(a) $\dfrac{e+2}{e^2-1}$

(b) $\dfrac{e^2+2}{e^2+1}$

(c) $\dfrac{1}{e-1}$

(d) $\dfrac{e+1}{e^2+1}$

(e) $\dfrac{1}{e+1}$

7. The area of the region bounded by the graph of $f(x) = x^3 - 2x^2 + 2x$ and its tangent line at the point $P(1, 1)$, is equal to

(a) $\frac{1}{4}$

(b) $\frac{1}{12}$

(c) $\frac{1}{8}$

(d) $\frac{1}{6}$

(e) $\frac{1}{2}$

8. If $f(x) = \int_x^{x^2-1} \sin\left(\frac{\pi}{2}t^2\right) dt$, then $f'(2) =$

(a) 1

(b) 4

(c) 2

(d) 3

(e) 0

9. Evaluate

$$\int_0^1 \frac{\sqrt{\arctan x}}{1+x^2}\, dx.$$

(a) $\frac{\pi\sqrt{\pi}}{6}$

(b) $\frac{\pi\sqrt{\pi}}{12}$

(c) $\frac{\pi\sqrt{\pi}}{2}$

(d) $\frac{\pi\sqrt{\pi}}{3}$

(e) $\frac{\pi\sqrt{\pi}}{4}$

10. A power series representation for $f(x) = \dfrac{1}{(5-x)^2}$ centered at $c = 2$ is

(a) $\sum_{n=0}^{\infty} 5^{1-n}(x-2)^n$

(b) $\sum_{n=0}^{\infty} 5^{2-n}(x-2)^n$

(c) $\sum_{n=0}^{\infty} \dfrac{n+1}{3^{n+2}}(x-2)^n$

(d) $\sum_{n=0}^{\infty} \dfrac{n}{3^{n+2}}(x-2)^n$

(e) $\sum_{n=0}^{\infty} \dfrac{n+2}{5^{n+2}}(x-2)^n$

11. A Maclaurin series for $\int_0^x \cos(t^2)\,dt$ is

(a) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n+1}}{(4n+1)(2n)!}$

(b) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{4n+1}}{(4n+1)(2n+1)!}$

(c) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{4n}}{(2n)!}$

(d) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n+1}}{(2n+1)(2n)!}$

(e) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{4n+1}}{(4n+1)(2n)!}$

12. The interval of convergence of $\sum_{n=2}^{\infty} \dfrac{(x-1)^n}{2^n\sqrt{n}}$ is

(a) $(-\infty, \infty)$

(b) $[-1, 3]$

(c) $[-1, 3)$

(d) $(-1, 3]$

(e) $(-1, 3)$

13. The series $\sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}\, n^2}{n^3+1}$

(a) diverges

(b) diverges by the Alternating series test

(c) converges absolutely

(d) converges conditionally

(e) converges absolutely by the Integral test

14. Consider the series

$$\sum_{n=1}^{\infty} \frac{4n^2+7}{n^{5/2}+5n+1}.$$

(a) The series converges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^{5/2}}$

(b) The series converges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$

(c) The series diverges by the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

(d) The series converges as $\lim_{n\to\infty} \frac{4n^2+7}{n^{5/2}+5n+1} = 0$

(e) The series converges by the comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$

15. The coefficient of x^2 in the nth degree Maclaurin polynomial of $f(x) = \frac{-x+\sin x}{x^3}$ is

(a) $-\frac{1}{5}$

(b) $\frac{1}{120}$

(c) $\frac{1}{20}$

(d) $-\frac{1}{6}$

(e) $-\frac{1}{80}$

16. Evaluate

$$\int_0^{\infty} \frac{dx}{\sqrt{x}\,(x+9)}.$$

(a) converges to $\frac{\pi}{3}$

(b) converges to π

(c) converges to $\frac{\pi}{9}$

(d) converges to 0

(e) diverges

17. The volume of the solid generated by revolving the region bounded by the graphs of $y = \dfrac{1}{x^2}$, $y = 0$, $x = 1$, and $x = 2$ about the y-axis, is equal to

(a) $\dfrac{\pi}{2} \ln 2$

(b) $\dfrac{3\pi}{2} \ln 2$

(c) $\pi \ln 2$

(d) $4\pi \ln 2$

(e) $2\pi \ln 2$

18. Which of the following series is/are convergent?

$$\text{I. } \sum_{n=1}^{\infty} \frac{99^n}{n!} \qquad \text{II. } \sum_{n=1}^{\infty} \left(\frac{1}{e} - e^{-n} \right)^n \qquad \text{III. } \sum_{n=2}^{\infty} \frac{n}{\ln n}$$

(a) I and II only

(b) I only

(c) II only

(d) II and III only

(e) III only

19. Which of the following series is/are convergent?

$$\text{I. } \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2} \qquad \text{II. } \sum_{n=1}^{\infty} \frac{1}{n^{2025/2026}} \qquad \text{III. } \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

(a) II only

(b) I and III only

(c) II and III only

(d) I and II only

(e) I only

20. Evaluate

$$\int_0^1 \frac{e^x - e^{-x}}{e^x + e^{-x}}\, dx.$$

(a) $\ln \dfrac{e^2+1}{2e}$

(b) $\ln \dfrac{e^2+1}{e^2-1}$

(c) $\ln \dfrac{e^2+1}{e}$

(d) $\ln \dfrac{e^2-1}{2e}$

(e) $\ln \dfrac{e^2-1}{e}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\frac{1+x-x^2}{x^2(1+x)} = \frac{1}{x^2} - \frac{1}{1+x}$; integral $= \frac{1}{2} - \ln\frac{3}{2}$.
2	**(b)**	$L = \sqrt{6+L} \Rightarrow L^2 - L - 6 = 0$, and $L \geq 0$, so $L = 3$.
3	**(c)**	$\lvert S - S_N\rvert \leq \frac{1}{(N+1)!} < 0.001$ needs $(N+1)! > 1000$: $7! = 5040$, so $N = 6$.
4	**(e)**	Infinite discontinuity at $x = 1$: $\int_0^1 \frac{dx}{(x-1)^2} = \infty$, so it diverges.
5	**(c)**	$u = \sqrt{x}$: $2\int_0^\pi u \sin u\, du = 2[-u\cos u + \sin u]_0^\pi = 2\pi$.
6	**(a)**	$\frac{1}{e-1} + \frac{1}{e^2-1} = \frac{e+2}{e^2-1}$ (two geometric series).
7	**(b)**	Tangent $y = x$; $f(x) - x = x(x-1)^2$; area $= \int_0^1 (x^3 - 2x^2 + x)\, dx = \frac{1}{12}$.
8	**(b)**	$f'(x) = 2x\sin(\frac{\pi}{2}(x^2-1)^2) - \sin(\frac{\pi}{2}x^2)$; $f'(2) = 4\cdot 1 - 0 = 4$.
9	**(b)**	$u = \arctan x$: $\int_0^{\pi/4} \sqrt{u}\, du = \frac{2}{3}(\frac{\pi}{4})^{3/2} = \frac{\pi\sqrt{\pi}}{12}$.
10	**(c)**	$\frac{1}{5-x} = \sum \frac{(x-2)^n}{3^{n+1}}$; differentiate: $\sum_{n=0}^\infty \frac{n+1}{3^{n+2}}(x-2)^n$.
11	**(e)**	$\cos t^2 = \sum(-1)^n \frac{t^{4n}}{(2n)!}$; integrate: $\sum(-1)^n \frac{x^{4n+1}}{(4n+1)(2n)!}$.
12	**(c)**	$R = 2$ about 1; $x = 3$ gives $\sum \frac{1}{\sqrt{n}}$ (diverges), $x = -1$ alternating (converges): $[-1, 3)$.
13	**(d)**	Alternating series test gives convergence; $\sum \lvert a_n\rvert \sim \sum \frac{1}{n}$ diverges: conditional.
14	**(c)**	Terms $\sim \frac{4}{\sqrt{n}}$; limit comparison with $\sum \frac{1}{\sqrt{n}}$ (ratio $\to 4$) gives divergence.
15	**(b)**	$f(x) = -\frac{1}{6} + \frac{x^2}{5!} - \cdots$, so the x^2 coefficient is $\frac{1}{120}$.
16	**(a)**	$u = \sqrt{x}$: $2\int_0^\infty \frac{du}{u^2+9} = \frac{2}{3}\cdot\frac{\pi}{2} = \frac{\pi}{3}$.
17	**(e)**	Shells: $V = 2\pi\int_1^2 x\cdot\frac{1}{x^2}\, dx = 2\pi\ln 2$.
18	**(a)**	I: ratio $\to 0$; II: root $\to \frac{1}{e} < 1$; III: terms $\to \infty$ diverges.
19	**(b)**	I: integral test gives $\frac{1}{\ln 2}$; II: $p < 1$ diverges; III telescopes to $\frac{1}{2}$.
20	**(a)**	$u = e^x + e^{-x}$: $[\ln(e^x + e^{-x})]_0^1 = \ln\frac{e^2+1}{2e}$.